

RONALD OSSAI

3BRIGHTON, UNITED KINGDOM · +447767693627

ronjjossai@gmail.com · [LinkedIn: Ronald Ossai](#) · [Portfolio: ronaldossai.xyz](#)

Summary –IT Specialist/Software Developer with strong background in data analysis, cloud infrastructure, and backend engineering. Hands-on experience in backend systems, scripting, and API integrations. 2+ years professional experience combining software development and IT helpdesk. Hands-on with Python, SQL, Power BI, and AWS. Experienced in building analytics tools, CI/CD pipelines, and scalable cloud applications.

PROFESSIONAL WORK EXPERIENCE

2020 (JULY-SEPTEMBER)

GRAFFITI FURNITURES LTD – INTERNSHIP

- Learned procedural and imperative programming language to help further understand the basic functions of the CNC machine.
- Learned about the functionality of rendering software for interior architecture.
- Facilitated logistical management tasks e.g. the movement of materials from shop floor to installation venue.
- Utilization of G-Code, M-Code to calculate dimensional drilling functions.

2021-2022 (SEPTEMBER-JUNE)

ICAEW LTD – STUDENT AMBASSADOR

- Social media brand handler in the University of Sussex
- Opening STEM opportunities for Ethnic minorities on the University Campus

2022-2023 (SEPTEMBER-SEPTEMBER)

ALLIANZ, SURREY – IT TRAINEE

- Junior Developer Working with Solutions Architect department for Allianz 2022/2023, 1-year full-time work experience.
- Utilized Docker for containerization of applications, streamlining development, testing, and deployment workflows.
- Contributed to DevOps and software development initiatives using Python, JavaScript, and SQL. Supported and maintained CI/CD pipelines, enhancing deployment efficiency and reliability. Gained hands-on experience with Amazon Web Services (AWS), supporting
- cloud-based infrastructure and deployment workflows.
- Designed and maintained data dashboards and reports using Power BI and Microsoft
- Excel, supporting internal decision-making and operations analytics.

2025 (JUNE-AUGUST)

CARTOGRAPHER IO, BRIGHTON – CONTRACT DEVELOPER

- Delegate Management System | React/TypeScript Full-Stack Application
- Developed a comprehensive administrative interface for managing RCA course delegates using React 18, TypeScript and Remix. Implemented complex CRUD operations with modal-based workflows for delegate assignment, group transfers and booking integrity management. Built responsive UI components with Tailwind CSS, integrated Drizzle ORM for database operations, and created dynamic filtering/search functionality. Solved challenging state management issues including dropdown menu controls, merge conflict resolution, and type safety implementations. Demonstrated expertise in modern React patterns, API integration and maintaining data consistency across related entities.

2025-PRESENT (DECEMBER- PRESENT)

KODYPAY LTD, LONDON – IT SPECIALIST

- Managed and configured core company IT systems including Google Workspace, Slack, and 1Password, ensuring secure and efficient day-to-day operations.
- Led employee onboarding and offboarding from a technical perspective, provisioning accounts, permissions, and devices while maintaining strict access control policies.
- Provided hands-on technical support across the organisation, troubleshooting hardware, software, and network issues and coordinating with external vendors when required.
- Implemented and enforced security best practices such as strong authentication policies, endpoint protection, and access management.
- Maintained full lifecycle management of company hardware inventory, including procurement, configuration, tracking, and asset auditing.
- Owned relationships with IT and office vendors, negotiating services and ensuring reliable and cost-effective support.
- Supported HR and operations with workplace setup, internal tooling, and technical logistics for company events and distributed teams.
- Continuously evaluated and introduced new tools and systems to improve productivity, collaboration, and organisational security.
- Administered the Twilio communications hub to support company-wide messaging workflows.
- Designed and implemented an AI-powered Slack bot that automatically summarises activity from critical Slack channels every 24 hours, improving operational visibility and reducing manual monitoring.

SKILLS

Languages & Frameworks: Python (Flask, Fast API, PyQt6), SQL, HTML/CSS, JavaScript, Java, C++, C#, C, Rust (novice)

Tools & Platforms: Git, Docker, AWS (basic), Linux, CI/CD, VSCode, GitHub

Concepts: OOP, REST APIs, Software Design Patterns, Agile Methodology, Unit Testing

EDUCATION

2021-2025

COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE BSc, UNIVERSITY OF SUSSEX

Honors in Computer Science and Artificial Intelligence BSc

Key modules included are:

Natural Language Engineering

Compilers and Computer Architecture

Machine learning

Computer Vision

Neural Networks

Quantum Computing.

2017-2019

FOUNDATION, BRIDGEHOUSE COLLEGE — INCLUDING A IT, A IN MATHEMATICS

2017

IGCSE, DAYWATERMAN COLLEGE NG- 2A*s (INCLUDING ICT), 3A's (INCLUDING MATHEMATICS AND PHYSICS) 3B's (INCLUDING ENGLISH LANGUAGE, CHEMISTRY AND FURTHER MATHEMATICS)

PROJECTS

This section contains a multitude of projects I have been able to work on over the years of consistent practicing and learning.

Stylia — Cross-Platform Mobile Application (React Native, TypeScript) • Developed a cross-platform mobile application using **React Native and Expo**, implementing multi-screen navigation, authentication flows, and persistent user data.

- Designed modular, reusable UI components and organised the codebase using a scalable folder structure with clear separation between screens, components, and logic.
- Implemented features for wardrobe management, outfit generation, calendar planning, and user surveys, demonstrating end-to-end mobile product architecture.
- Configured development tooling including TypeScript, ESLint, Prettier, and Husky pre-commit hooks to enforce code quality and maintain consistency across the codebase.

Interactive Standard Model — TypeScript, Three.js, React • Built a 3D visualization platform that renders subatomic particles (quarks, leptons, gauge bosons, and the Higgs boson) in a dynamic, real-time environment.

- Architected strict TypeScript interfaces to model particle properties, decay modes, quantum states, and interaction rules with total type safety.
- Engineered custom 3D web graphics utilizing Three.js, incorporating custom particle shaders, interactive camera controls, dynamic lighting, and orbital animations.
- Implemented raycasting and event-driven state management to allow user interactions like focusing, inspecting physics parameters, and triggering decay chain animations.
- Structured modular component hierarchies and optimized animation loops to guarantee 60 FPS performance across modern browsers.

Business Analytics Dashboard – Django, DRF, PostgreSQL, Chart.js • Built a secure, web-based analytics dashboard inspired by real-world use cases at Allianz.

- Implemented user authentication (JWT), CSV data ingestion, and background ETL tasks using Celery.
- Visualized KPIs using Chart.js and enabled PDF export of reports.
- Built a RESTful API with filtering, pagination, and role-based access control.
- Containerized with Docker and deployed to production on Render.
- Tech Stack: Django, Django REST Framework, PostgreSQL, Celery, Redis, Chart.js, Docker.

Particle Life Simulator – Python, Pygame Simulated complex particle interactions using a 6x6 matrix of behaviour rules. Implemented parameters for position, type, and physics, with $O(n^2)$ time complexity.

Toy Quantum Circuit Editor – Python, PyQt6, NumPy GUI-based editor for building quantum circuits with drag-and-drop functionality and OpenQASM 2.0 output. Simulated entanglement and Bell states.

WEB 3D: CTRL + LIFE – 3D website made using threejs, bootstrap, JavaScript and React

Binary Classification Model – Python, NumPy Built and trained a multilayer perceptron using backpropagation for a classification task. Achieved 83.5% accuracy on standardized data using 80/20 train-test split.

Computer Vision Alignment Tool – Python, OpenCV, NumPy

Used HOG descriptors and grayscale normalization to align facial features. Applied linear regression for image preprocessing.

REINFORCE Learning Agent – Python, OpenAI Gym

Evolved neural networks via genetic algorithms to control agents in simulation environments. Used fitness maximization to guide learning.

Inventory Management System – C#, ASP.NET Core, SQL Server

Role-based CRUD app with RESTful API, Docker deployment, and admin interface.

Unity Game: 2D Arena Shooter – C#, Unity

Implemented enemy waves, power-ups, and UI with Scriptable Objects.

Web Security Platform – PHP, JavaScript, MySQL

Built an antique valuation platform with multi-factor auth, brute-force protection, and secure file uploads.

Property Tycoon Game – Java

Java-based Monopoly clone using OOP principles, Entity management, and GUI.

INTEREST

Computational physics, systems programming, and music production.

Qualifications and Awards

- Bronze Duke of Edinburgh Award
- Microsoft Packages.
- BSc (Hons) Computer Science and Artificial Intelligence
- AWS Certified Solutions Architect

GITHUB AND PORTFOLIO:

Projects available at: github.com/ronaldossai

Portfolio: <https://ronaldossai.xyz>

LinkedIn: <https://www.linkedin.com/in/ronald-ossai-b66670206/>